

EnergyLens - Report Methodology at a Glance

What the 10-page Power BI report is built on - one-page executive reference.

Foundation

Microsoft Fabric medallion architecture (Bronze raw -> Silver clean -> Gold business logic) feeding a DirectLake semantic model (EnergyCopilotModel, ~213 measures). Gold is mirrored to Supabase mv_* for the web app.

The 10 pages

1 Portfolio	EUI benchmark, cost & CO2 totals - where the savings sit.
2 Building	EUI gauge, load profile, weather sensitivity, base load.
3 Anomalies	6 detection rules; spikes vs drift; estimated EUR impact.
4 Forecast & Recs	'do-nothing' forecast; measures scored on compliance + NPV + CO2 + payback, with an NPV viability gate.
5 Occupancy	base (empty) vs occupied load; ghost-load cost.
6 Sustainability	Scope 1/2/3; carbon intensity vs CRREM; EPC/MEPS.
7 HVAC	supply-return delta-T, runtime vs occupancy, COP/JAZ.
8 IoT	live power, CO2/comfort by zone, sensor uptime, FDD cost.
9 Battery	dispatch vs tariff, self-consumption, payback (Measured/Modeled/Prospect).
10 Solar	energy-basis PR, self-consumption vs export, specific yield, est. underperformance loss.

Key methodology

- PV PR is energy-basis ($\text{sum}(\text{gen}) / (\text{kWp} \times \text{sum}(\text{reference-yield}))$), clamped ≤ 0.95 , generation-weighted; $\text{PR} > 1$ is a data error, never real output.
- Recommendations: compliance-weighted score, but an $\text{NPV} \leq 0$ optional investment is gated to INFORMATIONAL (annuity 12 -> $\text{NPV} \leq 0 \Rightarrow \text{payback} \geq 12$ yr); compliance-mandated retrofits exempt.
- GHG Scope 2 uses a year-indexed grid factor (DE 0.363 location / 0.725 residual); carbon intensity tracked vs the CRREM stranding line.
- Thresholds are screening-grade, product-owner approved (2026-06-18); every modelled figure is labelled Est.

Data basis

Largely synthetic data with realistic structure. Real-telemetry exceptions are called out: the solar telemetry rollup (PR ~0.80) and three measured battery dispatches. Measured / Modeled / Prospect / Estimated tags keep real and modelled figures distinct.